Real Estate Investment Advisor

Predicting Property Profitability and Future Value

Project Report: Methodology, Findings and Use Cases

Labmentix Internship — Project 2

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October 2026

Live app: real-estate-advisor-stuti.streamlit.app

GitHub: stutigoyal8999/Labmentix_Internship_Business-Analytics

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1. Introduction and Problem Statement

Buying property is one of the largest financial decisions individuals and businesses make, yet buyers often rely on intuition or incomplete information. This project builds a machine learning tool that helps investors evaluate a property quickly and consistently by answering two questions:

- **Is this property a good investment?** A classification problem with a Yes/No answer and a confidence score.
- **What will this property be worth in 5 years?** A regression problem that estimates the future price in lakhs.

Objective

The objective was to clean and analyse a dataset of 250,000 Indian property listings, engineer meaningful features, train and compare classification and regression models, track every experiment with MLflow, and deliver the best models through an interactive, publicly deployed Streamlit application.

Deliverables

Deliverable	Where to find it
Cleaned and processed dataset	Project-2/data/cleaned_housing_data.csv
Python code for EDA, model training and evaluation	notebooks/preprocessing.ipynb, notebooks/exploratory_data_analysis.ipynb, train.py
MLflow experiment logs and registered models	Tracked locally with MLflow; evidence in Section 8 and the screenshots folder
Streamlit application with predictions and insights	app.py, deployed at real-estate-advisor-stuti.streamlit.app
Project documentation	This report and Project-2/README.md

2. Dataset Overview

The dataset contains **250,000 property records** and **23 columns**, covering **20 states, 42 cities and 500 locality names**. Each row describes one property listing.

Group	Columns
Location	State, City, Locality
Property	Property_Type, BHK, Size_in_SqFt, Year_Built, Age_of_Property, Floor_No, Total_Floors, Furnished_Status, Facing
Price	Price_in_Lakhs, Price_per_SqFt
Neighbourhood	Nearby_Schools, Nearby_Hospitals, Public_Transport_Accessibility
Facilities	Parking_Space, Security, Amenities
Listing	Owner_Type, Availability_Status, ID

Prices range from ₹10 to ₹500 lakhs and sizes from 500 to 5,000 sq ft. The dataset description mentions a crime rate feature and a "Sold" availability status, but **neither appears in the data**. This affects one EDA item in the brief (Section 6.3).

3. Tools and Libraries Used

Tool / Library	Purpose in this project
Python 3.11 (Anaconda environment)	Dedicated environment so library versions do not conflict with other projects
VS Code with Jupyter	Notebooks for preprocessing and EDA; Python scripts for training and the app
Pandas, NumPy	Data loading, cleaning, feature engineering and aggregation
Matplotlib, Seaborn	EDA charts and in-app visualisations
Scikit-learn	Pipelines, scaling, one-hot encoding, Logistic/Linear Regression, Random Forest, evaluation metrics
XGBoost	Gradient boosting models (best for both tasks)
MLflow	Experiment tracking, artifact logging and Model Registry
Streamlit, Streamlit Community Cloud	Interactive web application and free public deployment
Git and GitHub	Version control and hosting; the app deploys directly from the repository

4. Data Preprocessing

The data was assessed for quality issues before any analysis. Each issue and the decision taken is summarised below.

Check / Issue	Finding	Action and reasoning
Missing values	None in any column	No imputation needed
Duplicate rows	None, including when ID is ignored	No action needed
Price_per_SqFt units	Stored in lakhs and rounded to 2 decimals (e.g. 0.10 instead of ₹10,332); some values were 0.0	Recalculated in rupees as Price × 100,000 ÷ Size for full precision
Floor_No > Total_Floors	116,304 rows (47%) had impossible floor numbers	Capped Floor_No at Total_Floors and added a Floor_Inconsistent flag; deleting 47% of the data was not acceptable
Age_of_Property	Equal to 2025 − Year_Built in every row	Consistent; Year_Built dropped from modelling as redundant
Amenities	Multiple values in one cell, 325 combinations	Converted to Amenity_Count plus one 0/1 column per amenity
Outliers	19,723 properties (7.9%) above the IQR upper limit for price per sq ft (₹32,764); none for size	Kept: they are mathematically valid (small, expensive homes), and tree models are robust to outliers

4.1 Feature Engineering

- **Amenity_Count** and **Has_Gym, Has_Pool, Has_Clubhouse, Has_Garden, Has_Playground**.
- **School_Density_Score:** nearby schools scaled to 0–1.
- **Infrastructure_Score:** average of schools, hospitals and transport access (Low = 2, Medium = 6, High = 10), on a 1–10 scale.

4.2 Encoding and Scaling

Numeric features are standardised and categorical features one-hot encoded inside a scikit-learn **Pipeline** for each model. Fitting these steps on the training data only prevents information from the test set leaking into the model. The final cleaned dataset has **250,000 rows and 36 columns**.

5. Target Variable Design

The dataset does not include either target, so both were constructed using the approaches suggested in the brief. Because EDA showed that features are unrelated to price (Section 6), feature-based rules were chosen so the targets reflect sensible investment logic.

5.1 Regression Target: Future Price after 5 Years

Future Price = Current Price × $(1 + r)^5$, where the yearly growth rate r starts at **8%** and is adjusted by property features:

Factor	Adjustment	Reasoning
Public transport access	High +1%, Low −1%	Connectivity drives long-term demand
Property age	≤ 10 years +1%, > 25 years −1%	Newer properties hold value better
Amenities	4 or more: +0.5%	Well-equipped properties attract buyers
Security	Available: +0.5%	Gated, secure properties command premiums

This gives yearly growth rates between **6% and 11%** (mean 8.4%).

5.2 Classification Target: Good Investment

A multi-factor score awards one point for each condition met. A property scoring **3 or more out of 5** is labelled a Good Investment.

- Price per sq ft at or below the **city median** (good value relative to its market)
- **3 or more BHK** (wider buyer and rental demand)
- **Ready to move** (no construction risk)
- **Medium or High** public transport access
- **5 or more** nearby schools

This labels **63.6%** of properties as Good and 36.4% as Not Good, a mild imbalance handled by reporting the F1-score alongside accuracy.

6. Exploratory Data Analysis

All 20 EDA questions in the brief were answered with charts in *exploratory_data_analysis.ipynb*. The central finding is that the dataset is **synthetic**: prices, sizes and categories are spread almost perfectly evenly, and features were generated independently of price.

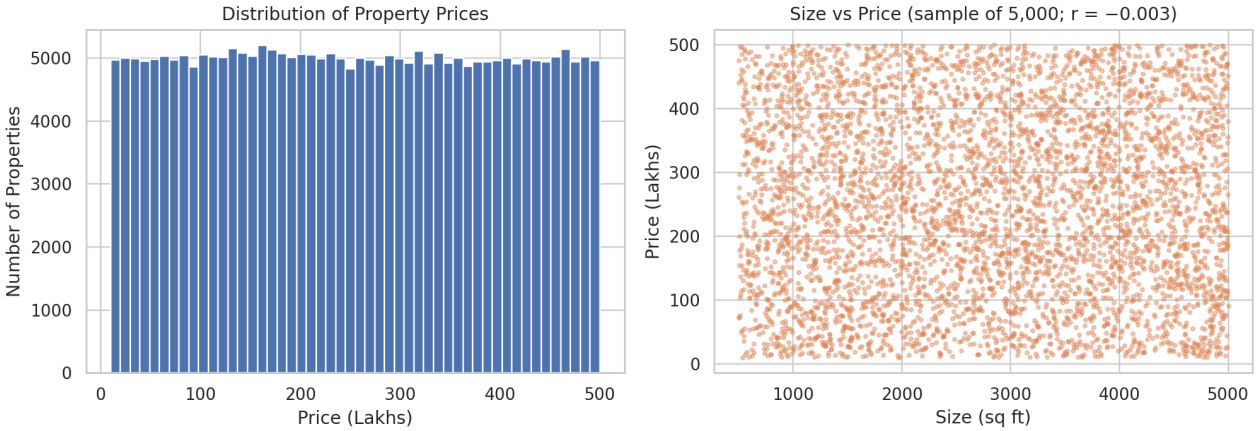


Figure 1: Left: prices are uniformly distributed between ₹10 and ₹500 lakhs. Right: size and price are unrelated (correlation −0.003).

6.1 Answers to the 20 EDA Questions

#	Question	Finding
1	Distribution of prices	Uniform from ₹10–500 lakhs; mean ₹254.6 lakhs, skewness 0.01
2	Distribution of sizes	Uniform from 500–5,000 sq ft; median 2,747 sq ft
3	Price per sq ft by property type	Nearly identical: median ₹9,190–9,305 for all three types
4	Size vs price	No relationship (r = −0.003)
5	Outliers	Price per sq ft: 19,723 (7.9%) above ₹32,764, kept as valid. Size: none
6	Price per sq ft by state	₹12,931 (Punjab) to ₹13,252 (Karnataka): only 2.5% apart
7	Average price by city	₹250.79 (Cuttack) to ₹258.46 lakhs (Bangalore): only 3.1% apart
8	Median property age by locality	16–21 years; most localities 18–19. Locality names repeat across all 42 cities
9	BHK across cities	Each BHK (1–5) is about 20% in every city (18.6–22.0%)
10	Top 5 localities' price trends	Prices fluctuate between ₹245–309 lakhs with no consistent trend
11	Correlations	Only between calculated columns (e.g. price and future price 0.99); all others near 0
12	Schools vs price per sq ft	No effect: medians ₹9,144–9,352; r ≈ 0
13	Hospitals vs price per sq ft	No effect: medians ₹9,129–9,349; r ≈ 0
14	Price by furnished status	About ₹254–255 lakhs for all three statuses
15	Price per sq ft by facing	Within 1%; West slightly higher (₹9,308)
16	Owner types	Broker, Owner and Builder about one-third each
17	Availability status	Exactly 50/50 Ready to Move vs Under Construction; no "Sold"

#	Question	Finding
18	Parking vs price	₹254.75 vs ₹254.43 lakhs: no meaningful effect
19	Amenities vs price per sq ft	Differences under 0.5% by count or individual amenity
20	Transport vs price / investment	No effect on price; Medium/High access has 76% Good Investment rate vs 39% for Low (by target design)

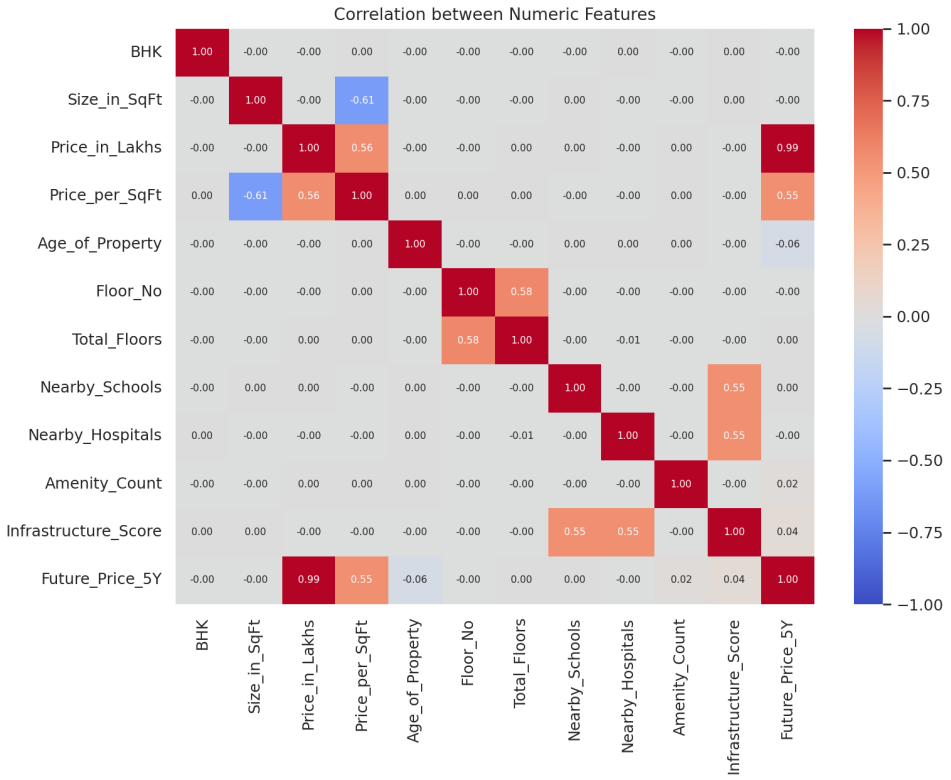


Figure 2: Correlation matrix. Strong values appear only between columns derived from each other.

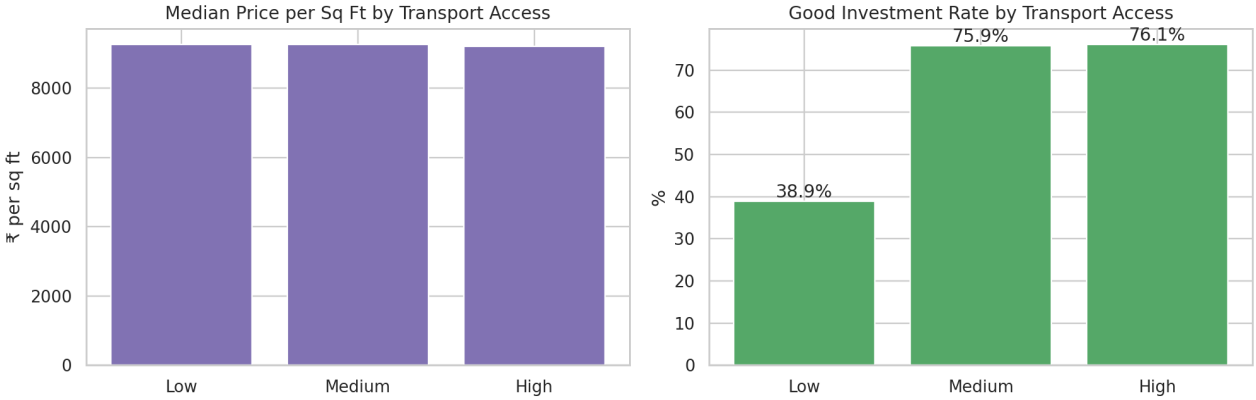


Figure 3: Transport access does not change price per sq ft, but strongly affects investment potential through the target rules.

6.2 Additional Analyses Requested in the Brief

Analysis	Finding
Price trends by city	Average prices differ by only ₹7.7 lakhs (3.1%) across all 42 cities (Q7)

Analysis	Finding
Area vs investment return	Size has no correlation with 5-year growth (r = 0.004). It has a weak positive link with Good Investment (r = 0.18), because larger homes have lower price per sq ft and so pass the "below city median" rule more often
Infrastructure score vs resale value	Weak link with future price (r = 0.04) but a moderate link with growth rate (r = 0.44), driven by transport access in the growth rule
Impact of crime rate	**Could not be analysed:** the dataset contains no crime rate column. It is listed as future work (Section 13)

7. Model Training and Evaluation

Training is done in *train.py*. The data was split **80% training / 20% test** (50,000 test properties), stratified by class for classification. Each model is a pipeline of preprocessing plus estimator. 26 features were used; ID, State, Locality, Year_Built, the raw Amenities text, the cleaning flag and the intermediate target columns (Growth_Rate, Investment_Score) were excluded.

7.1 Classification: Good Investment

Model	Key settings	Accuracy	F1-score
Logistic Regression	max_iter = 1000	0.9015	0.9230
Random Forest	100 trees, max depth 15	0.9986	0.9989
XGBoost (best)	300 trees, depth 6, learning rate 0.1	**0.9995**	**0.9996**

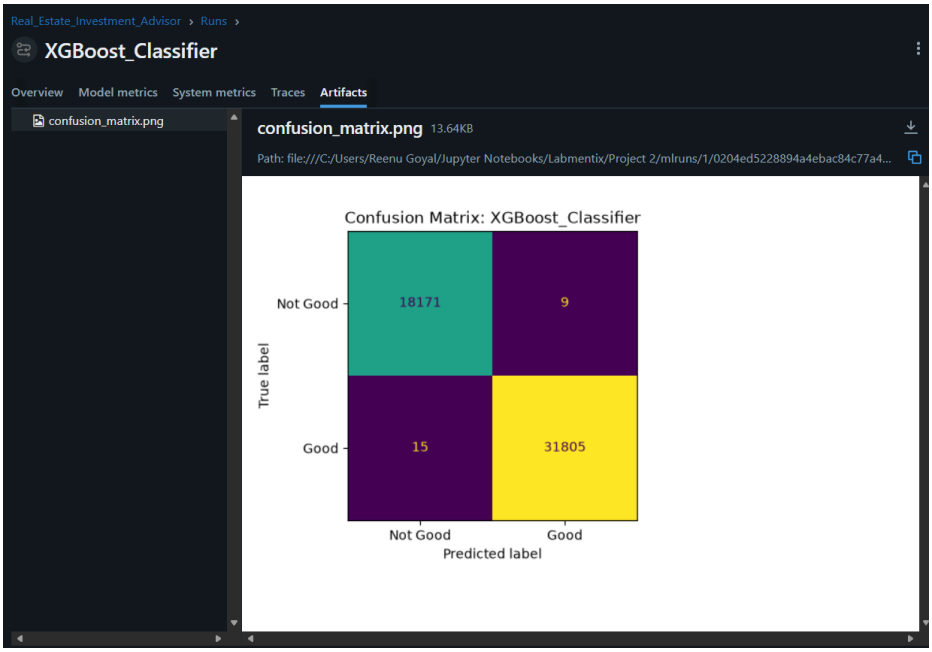


Figure 4: Confusion matrix of the XGBoost classifier: only 24 errors (9 false positives, 15 false negatives) out of 50,000 test properties.

7.2 Regression: Future Price after 5 Years

Model	Key settings	RMSE (lakhs)	MAE (lakhs)	R^2
Linear Regression	Default	12.55	9.48	0.99654
Random Forest	50 trees, depth 12, min leaf 20	1.25	0.96	0.99997

Model	Key settings	RMSE (lakhs)	MAE (lakhs)	R^2
XGBoost (best)	500 trees, depth 6, learning rate 0.1	**0.98**	**0.81**	**0.99998**

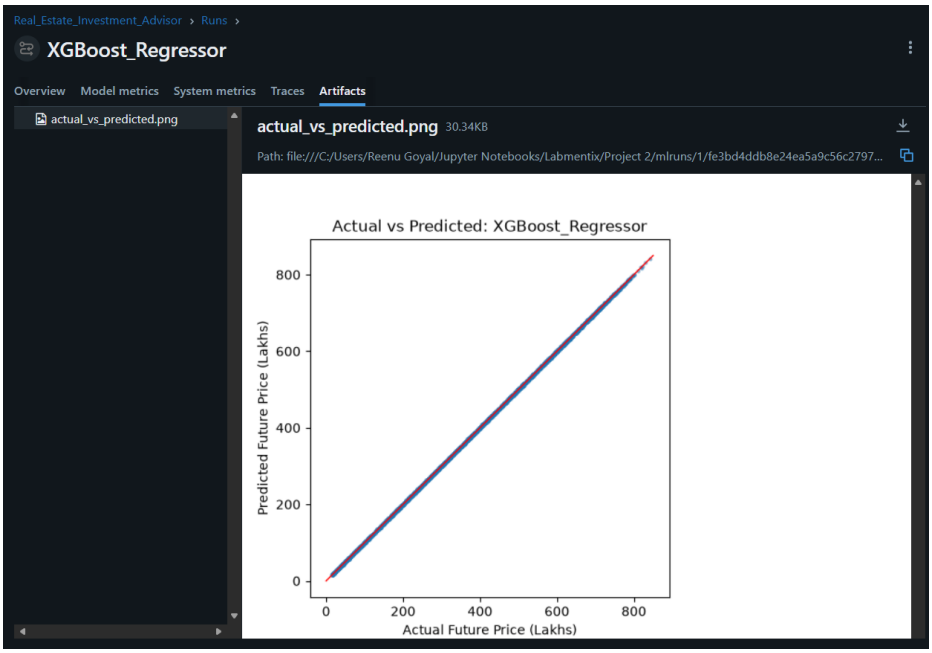


Figure 5: Actual vs predicted future price for XGBoost. Points lie on the diagonal; the average error is under 0.3% of the mean future price (₹382 lakhs).

7.3 Interpreting the Results

The near-perfect scores are expected and should not be read as real-world market accuracy. Because both targets were built from rules on the input features, the models are learning those rules. The comparison is still informative: the **linear baselines** score noticeably lower because they cannot represent thresholds ("3 or more BHK") or multiplication (price × growth), while **tree-based models** capture both. Feature importance (Section 9) confirms each model relies on exactly the features used in its target's rules.

8. Experiment Tracking with MLflow

Every model run logged its parameters, metrics, a diagnostic chart and the trained model to MLflow (SQLite backend). The best model for each task was registered in the **MLflow Model Registry**. The MLflow run files are not stored on GitHub because they exceed its size limits, so the screenshots below serve as the record.

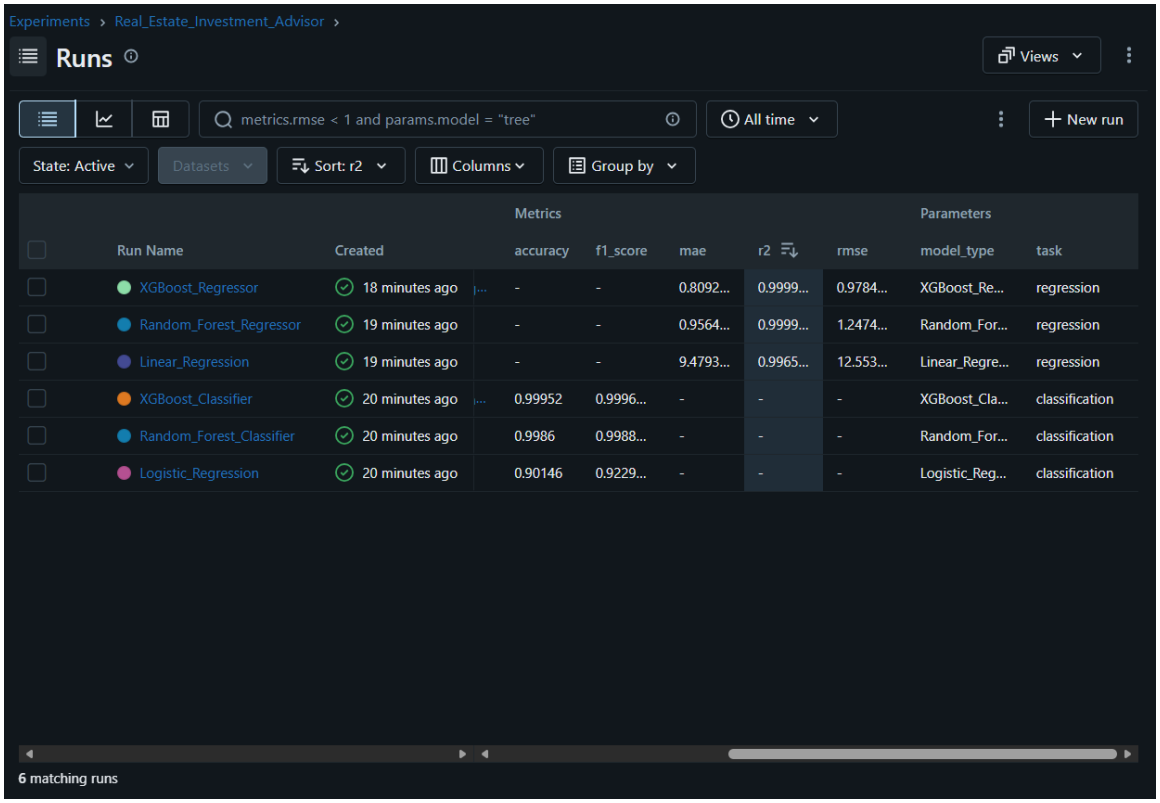
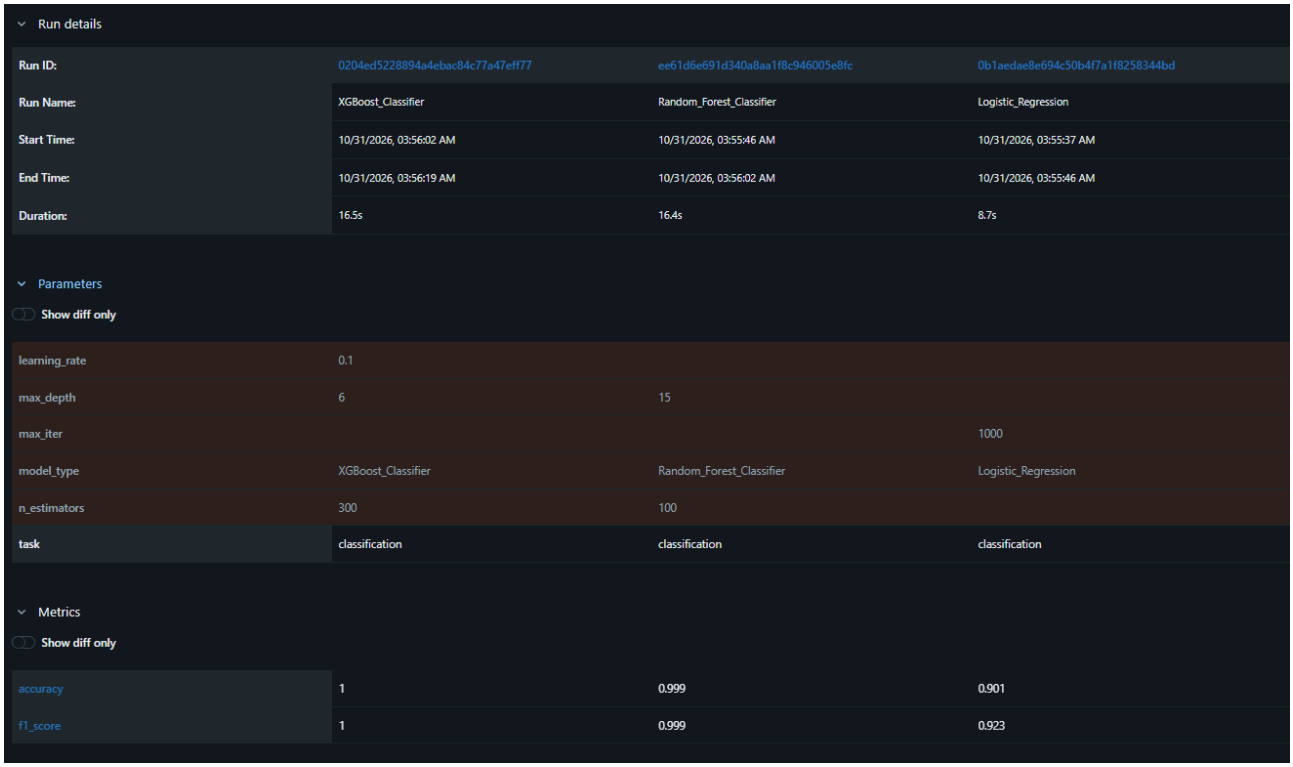


Figure 6: All six runs in the Real_Estate_Investment_Advisor experiment, with metrics and parameters.



Run details

Run ID:	0204ed5228894a4ebac84c77a47eff77	ee61d6e691d340a8aa1f8c946005e8fc	0b1aedae8e694c50b4f7a1f8258344bd
Run Name:	XGBoost_Classifier	Random_Forest_Classifier	Logistic_Regression
Start Time:	10/31/2026, 03:56:02 AM	10/31/2026, 03:55:46 AM	10/31/2026, 03:55:37 AM
End Time:	10/31/2026, 03:56:19 AM	10/31/2026, 03:56:02 AM	10/31/2026, 03:55:46 AM
Duration:	16.5s	16.4s	8.7s

Parameters

Show diff only

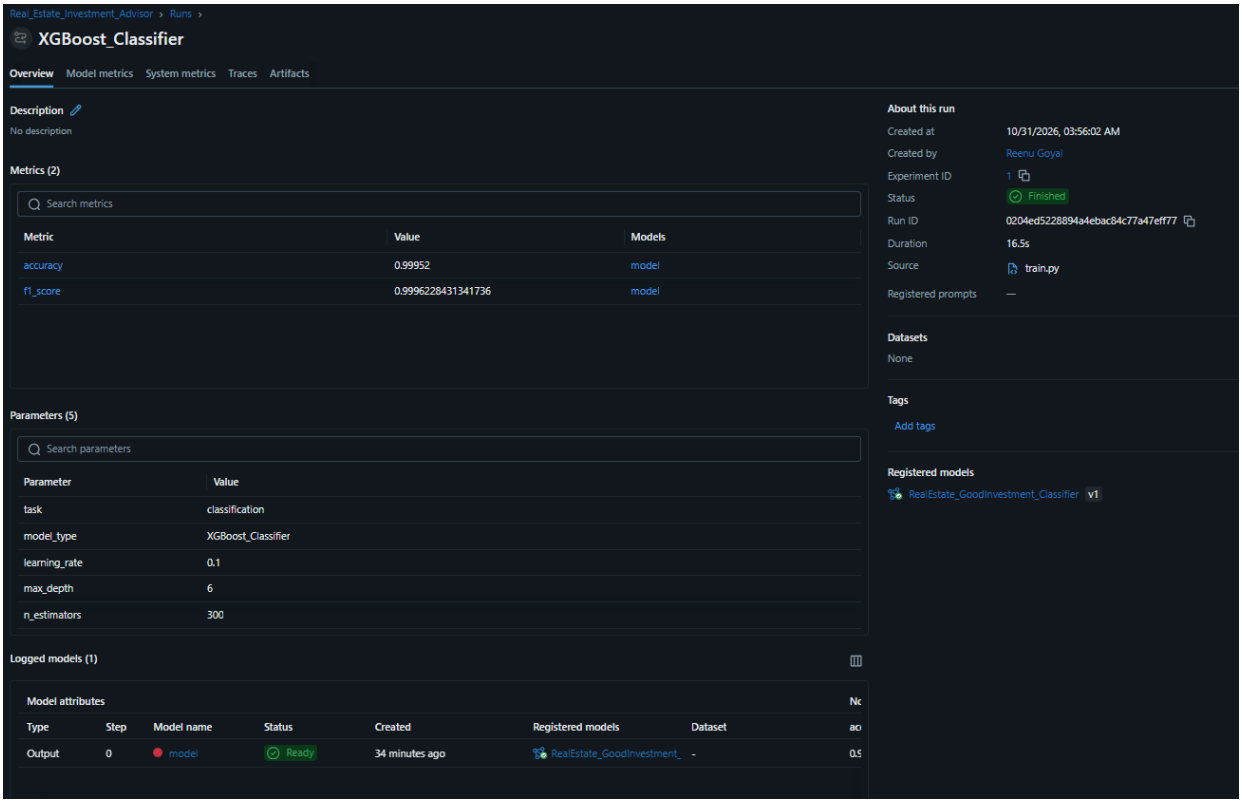
learning_rate	0.1		
max_depth	6	15	
max_iter			1000
model_type	XGBoost_Classifier	Random_Forest_Classifier	Logistic_Regression
n_estimators	300	100	
task	classification	classification	classification

Metrics

Show diff only

accuracy	1	0.999	0.901
f1_score	1	0.999	0.923

Figure 7: Side-by-side comparison of the three classifiers. MLflow rounds to three decimals here, so XGBoost's 0.9995 is shown as 1.



Real_Estate_Investment_Advisor › Runs ›

XGBoost_Classifier

Overview | Model metrics | System metrics | Traces | Artifacts

Description

No description

Metrics (2)

Search metrics

Metric	Value	Models
accuracy	0.99952	model
f1_score	0.999628431341736	model

Parameters (5)

Search parameters

Parameter	Value
task	classification
model_type	XGBoost_Classifier
learning_rate	0.1
max_depth	6
n_estimators	300

Logged models (1)

Model attributes							Nc
Type	Step	Model name	Status	Created	Registered models	Dataset	ac
Output	0	model	Ready	34 minutes ago	RealEstate_GoodInvestment_	-	0.9

About this run

Created at	10/31/2026, 03:56:02 AM
Created by	Reenu Goyal
Experiment ID	1
Status	Finished
Run ID	0204ed5228894a4ebac84c77a47eff77
Duration	16.5s
Source	train.py
Registered prompts	—

Datasets

None

Tags

Add tags

Registered models

RealEstate_GoodInvestment_Classifier v1

Figure 8: XGBoost classifier run, linked to the registered model RealEstate_GoodInvestment_Classifier v1.

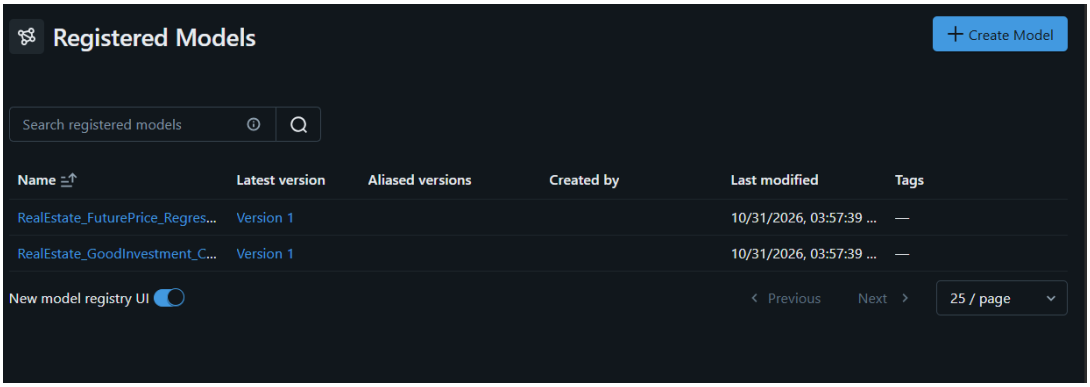


Figure 9: Both best models registered in the MLflow Model Registry as Version 1.

9. Streamlit Application and Deployment

The app (*app.py*) loads the two registered models and the cleaned dataset, and is deployed on Streamlit Community Cloud at real-estate-advisor-stuti.streamlit.app. It has four tabs:

Tab	Features
Predict	Form for all property details; Good Investment verdict with model confidence; estimated price after 5 years with total and yearly growth; breakdown of the five investment criteria
Explore Properties	Filters by city, property type, price, size, BHK and Good Investment; summary metrics and results table
Market Insights	Location-wise heatmap (city × property type) with a choice of four metrics; price trend by construction year; top 10 cities by Good Investment rate
Model Insights	Evaluation metrics, comparison of all six models, confusion matrix, feature importance for both models, explanation of the target rules

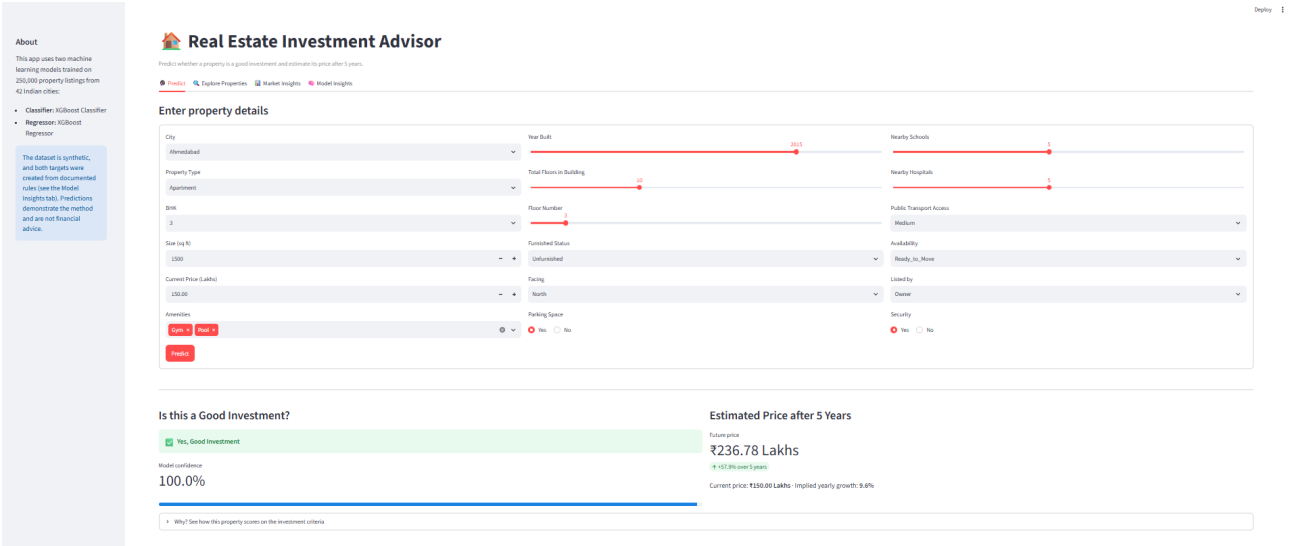


Figure 10: Prediction for a 3 BHK, 1,500 sq ft Ahmedabad apartment priced at ₹150 lakhs: Good Investment (100% confidence), estimated ₹236.78 lakhs in 5 years.

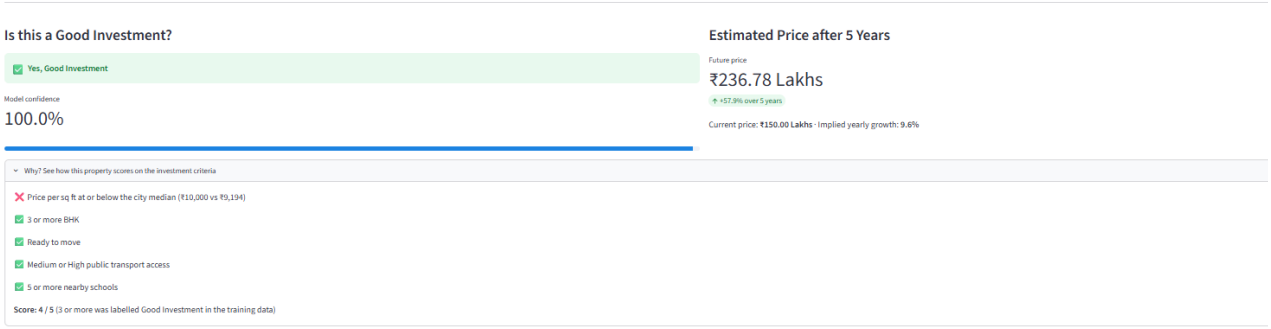


Figure 11: The explanation panel: the property meets 4 of 5 criteria (score 4/5).

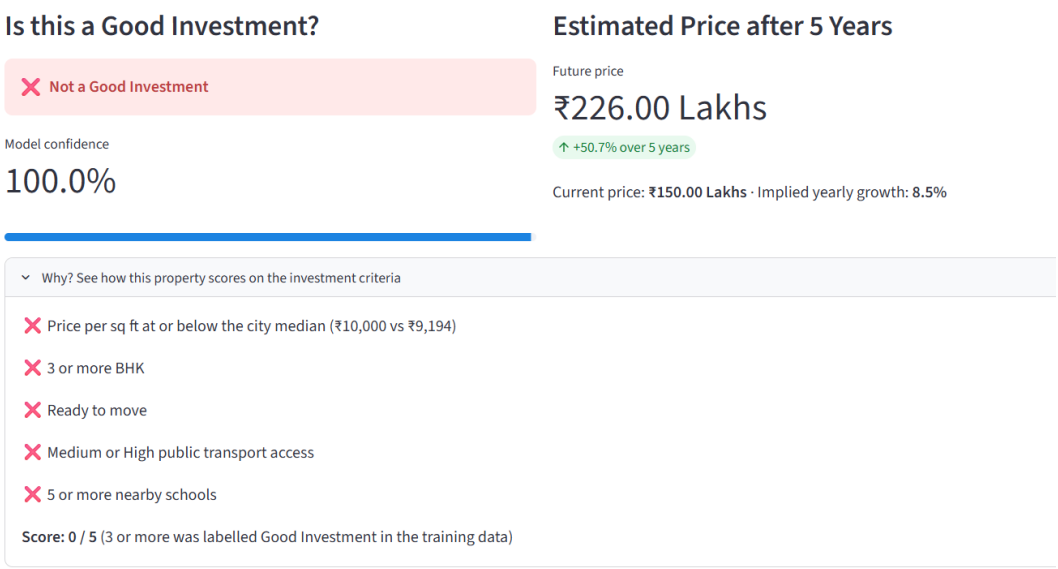
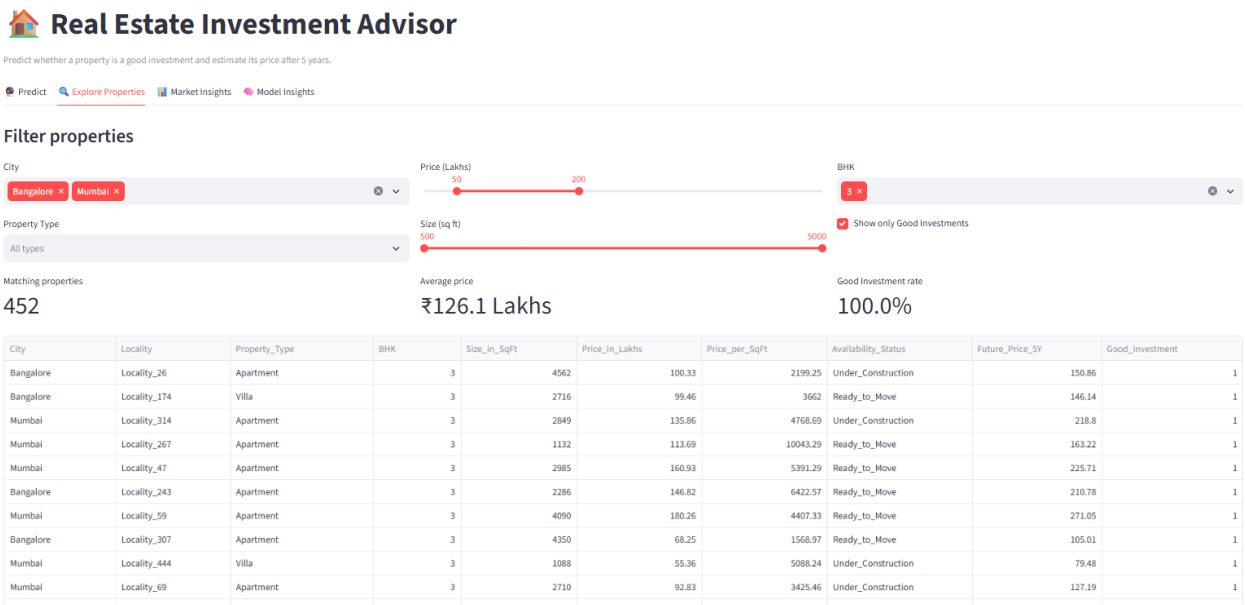


Figure 12: Changing to 2 BHK, 3 schools, Low transport and Under Construction gives Not a Good Investment (score 0/5) and lower growth (8.5% a year, ₹226.00 lakhs).



City	Locality	Property_Type	BHK	Size_in_SqFt	Price_in_Lakhs	Price_per_SqFt	Availability_Status	Future_Price_5Y	Good_Investment
Bangalore	Locality_26	Apartment	3	4562	100.33	2199.25	Under_Construction	150.86	1
Bangalore	Locality_174	Villa	3	2716	99.46	3662	Ready_to_Move	146.14	1
Mumbai	Locality_314	Apartment	3	2849	135.86	4768.69	Under_Construction	218.8	1
Mumbai	Locality_267	Apartment	3	1132	113.69	10043.29	Ready_to_Move	163.22	1
Mumbai	Locality_47	Apartment	3	2985	160.93	5391.29	Ready_to_Move	225.71	1
Bangalore	Locality_243	Apartment	3	2286	146.82	6422.57	Ready_to_Move	210.78	1
Mumbai	Locality_59	Apartment	3	4090	180.26	4407.33	Ready_to_Move	271.05	1
Bangalore	Locality_307	Apartment	3	4350	68.25	1568.97	Ready_to_Move	105.01	1
Mumbai	Locality_444	Villa	3	1088	55.36	5088.24	Under_Construction	79.48	1
Mumbai	Locality_69	Apartment	3	2710	92.83	3425.46	Under_Construction	127.19	1

Figure 13: Filtering 3 BHK Good Investments in Bangalore and Mumbai priced ₹50–200 lakhs returns 452 properties.

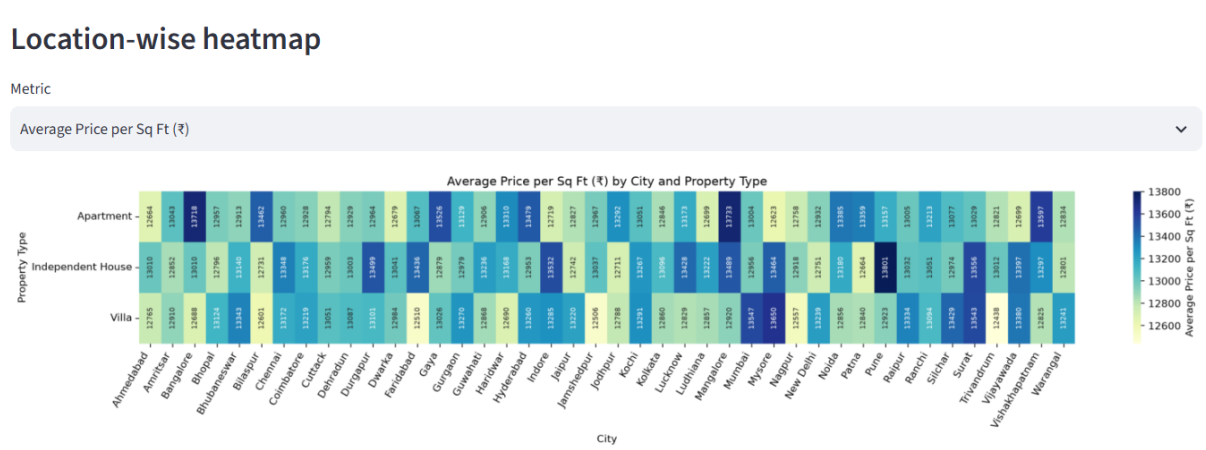


Figure 14: Location-wise heatmap of average price per sq ft by city and property type.

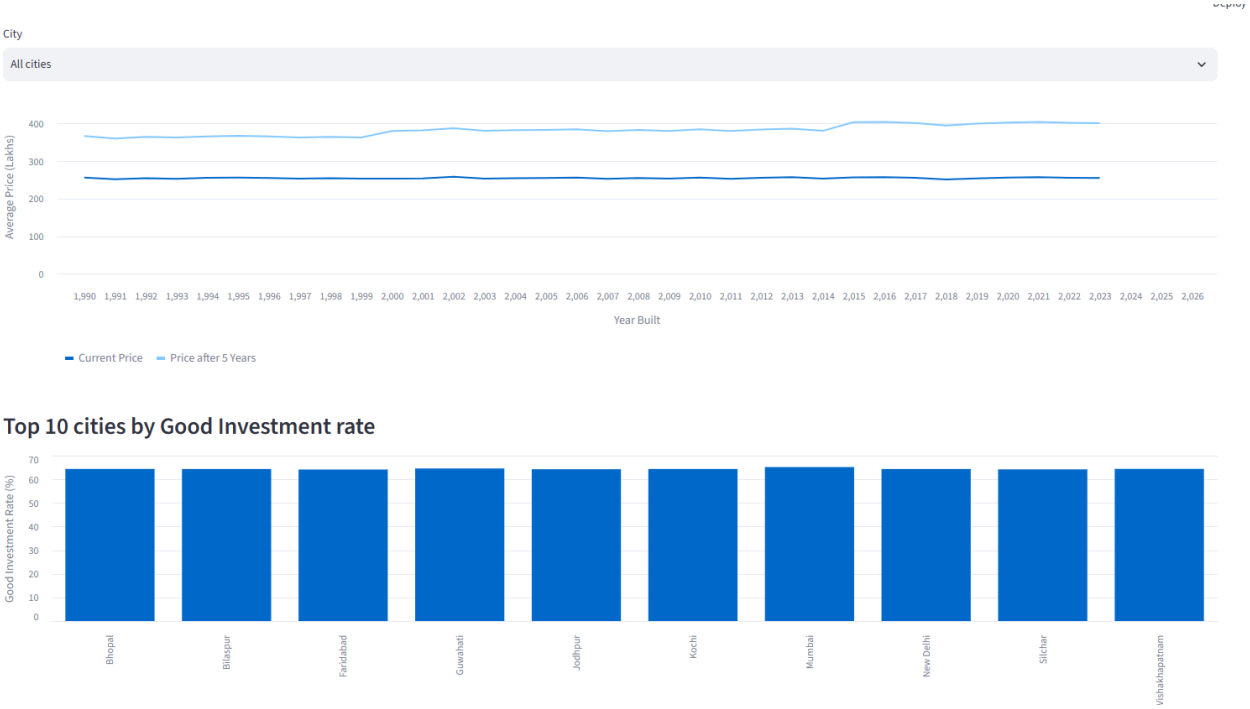


Figure 15: Current price is flat across construction years, while the 5-year price steps up for properties built from 2000 and from 2015 (age-based growth adjustments).

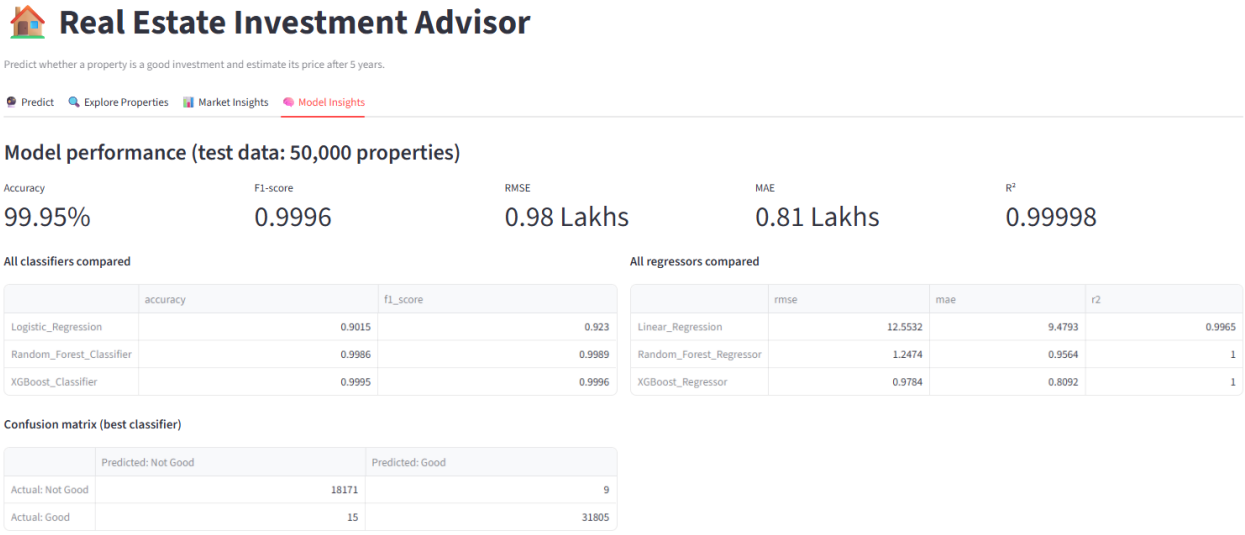


Figure 16: Model Insights tab: metrics, model comparison and confusion matrix.

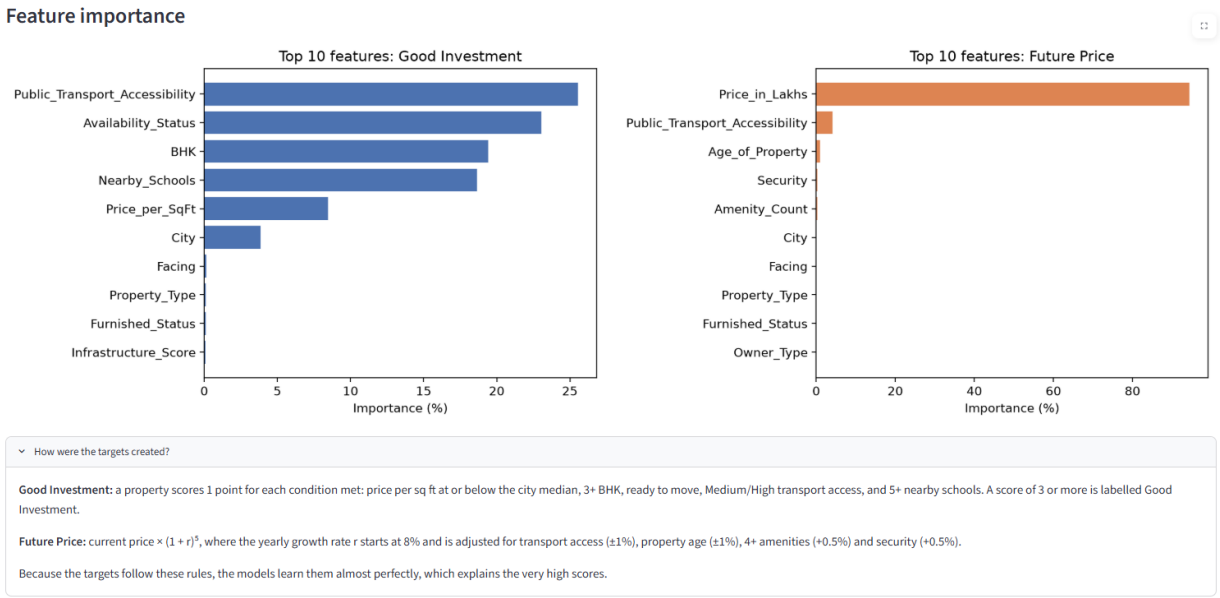


Figure 17: Feature importance. The classifier relies on transport access, availability, BHK, schools and price per sq ft, the five investment criteria. The regressor is driven by current price (about 94%), followed by the growth-rate factors.

10. Summary of Key Findings

- **The data is synthetic.** Prices, sizes, BHK, owner types and availability are spread almost perfectly evenly, and no property feature, location or amenity meaningfully affects price.
- **Location has almost no effect:** prices differ by only 2.5% across states and 3.1% across cities.
- **Data quality issues were found and fixed:** price per sq ft stored in the wrong unit, and impossible floor numbers in 47% of rows.
- **Rule-based targets were necessary** and were designed around sensible investment logic (value, size, readiness, connectivity, schools).

- **XGBoost was best for both tasks:** 99.95% accuracy (F1 0.9996) and RMSE of ₹0.98 lakhs (R^2 0.99998).
- **Feature importance matches the target rules,** confirming the models learned the intended logic rather than noise.
- **Transport access matters most for investment potential:** properties with Medium/High access are about twice as likely to be Good Investments (76% vs 39%).

11. Use Cases

User	How the tool helps
Home buyers and individual investors	Check in seconds whether a listing is good value and estimate its future worth before negotiating or committing
Real estate agents and advisors	Back recommendations to clients with consistent, explainable criteria and a projected 5-year value
Developers and analysts	Compare cities and property types on price per sq ft, growth and investment rate to spot where good-value properties are concentrated
Banks and lenders	Use projected value and investment score as one input when assessing property-backed loans
Template for real data	The full pipeline (cleaning, feature engineering, tracked training, deployment) can be re-run on real market data by replacing the dataset and target rules

12. Challenges and Solutions

Challenge	Solution implemented
Price_per_SqFt was stored in lakhs and rounded, losing precision	Recalculated it in rupees from price and size
47% of rows had a floor number above the building's total floors	Capped values and added a flag column instead of deleting almost half the data
Features showed no relationship with price, so real market patterns could not be learned	Designed documented, rule-based targets and stated clearly that high scores reflect learned rules, not market accuracy
Risk of data leakage from preprocessing	Placed scaling and encoding inside scikit-learn pipelines fitted on training data only
Re-running a data-modifying notebook cell recalculated the floor flag as all zeros	Re-ran the notebook from the top (Restart → Run All) and adopted this as standard practice
New Streamlit app required separate .py scripts, while EDA suited notebooks	Used VS Code with a dedicated Anaconda environment for both notebooks and scripts
MLflow 3 refused to save tree models in its default secure (skops) format	Switched to the cloudpickle format for these self-trained, trusted models
MLflow run files were too large for GitHub	Excluded them with .gitignore and documented the experiments with screenshots
Heatmap of 42 cities did not fit on screen in the wide app layout	Transposed it so cities run across and property types down
Deployment failed: the folder name "Project 2" contained a space, which broke the installer's path	Renamed the folders to Project-1 and Project-2; worked around a Windows file lock by copying the folder and updating Git tracking

Challenge	Solution implemented
Saved models must load with the same library versions online	Pinned exact versions of every library in requirements.txt and set Python 3.11 on Streamlit Cloud
Relative file paths broke when the app ran from the repository root	Located data and model files relative to app.py itself

13. Limitations and Future Work

- **Synthetic data:** predictions demonstrate the method and are not financial advice. The next step is to apply the pipeline to real listings, for example from property portals or government registration data.
- **Rule-based targets:** real future prices and investment outcomes would allow the models to learn genuine market behaviour rather than predefined rules.
- **Missing features:** crime rate, RERA registration and actual transaction history were unavailable; adding them would enable the crime-rate analysis in the brief.
- **Locality names** repeat across cities; real neighbourhood identifiers would enable meaningful local analysis.
- **App enhancements:** loading models directly from the MLflow registry, comparing multiple properties side by side, and adding map-based visualisations.

14. Conclusion

This project delivered a complete, end-to-end machine learning solution: data cleaning that corrected two significant quality issues, a thorough EDA answering all 20 questions, carefully designed targets, six tracked and compared models with the best two registered in MLflow, and a publicly deployed Streamlit app that turns predictions into clear, explainable investment guidance.

A key outcome of the analysis was recognising that the dataset is synthetic and that its features do not drive price. Rather than presenting near-perfect scores as real-world accuracy, the report documents how the targets were built and why the models perform as they do. The same pipeline is ready to be applied to real market data, where it could provide genuine value to buyers, investors and advisors.